Exercise 2 — Step-by-Step Solutions

Advanced System Theory · System Properties and Linearization

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan) **Covers:** Task 1 (a–d), Task 2 (a, b and the I/O-relation part, called "2c" here because the sheet labels it "b" twice), Task 3 (a–d) **Checking:** every result has been cross-checked against the official *Solution to Exercise 2* (RAT) and agrees with it; notes marked **"Official solution:"** point out where the official sheet presents something differently. Every result is also verified by the scripts in `python/` (symbolic algebra with SymPy, numerical search, simulation).

Every step is written out. References like "Concepts §3.2" point to `concepts.md`. The notation is that of the lecture slides: bold symbols are vectors and matrices ($\boldsymbol{x}$, $\boldsymbol{f}$, $\boldsymbol{A}$, ...), $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ is the equilibrium, $\Delta\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$, and $\cdot|_0$ means evaluation at the equilibrium. Symbols given on the exercise sheet (ϕ, $(\boldsymbol{x}^*, u^*)$, $\delta[k]$, ...) are kept as they are.

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Task 1 — System properties

For each I/O relationship we test the three definitions of the lecture:

- **Causality (Definition 1.3):** $u_1(t) = u_2(t)\ \forall t \le T \ \Rightarrow\ y_1(t) = y_2(t)\ \forall t \le T$, for every $T > 0$ (Concepts §4.2).
- **Time-invariance (Definition 1.2):** $y(u(t), t - T) = y(u(t - T), t)$ for every $T > 0$. The I/O relationships have no initial state, so $\boldsymbol{x}_0$ is dropped. Left side: output delayed by T (replace every t by $t - T$). Right side: output for the delayed input $u(t - T)$ (Concepts §3.2).
- **Linearity (Definition 1.1 / slide 12):** $u_1(t) \to y_1(t),\ u_2(t) \to y_2(t) \Rightarrow \alpha_1 u_1(t) + \alpha_2 u_2(t) \to \alpha_1 y_1(t) + \alpha_2 y_2(t)$ (Concepts §2.3).

A "Y" is proved in general; an "N" is shown with one counterexample (Concepts §1.3). Throughout, $A \neq 0$ and $f_0 \neq 0$ (special cases are noted at the end).

1(a) $y(t) = u(t) - 0.5u(t+1)$

Causality. The output at time t reads the input at t and at $t + 1$. The read time $t + 1 > t$ lies in the future for every t.

Counterexample. Take any $T > 0$. Let $u_1(t) = 0$ for all t, and $u_2(t) = 0$ for $t \le T$, $u_2(t) = 1$ for $t > T$. Then $u_1(t) = u_2(t)\ \forall t \le T$. At $t = T$: $y_1(T) = 0$, but $y_2(T) = u_2(T) - 0.5u_2(T+1) = 0 - 0.5 = -0.5$. So $y_1(t) \neq y_2(t)$ for a $t \le T$ ⇒ **not causal**.

Time-invariance.

- Right side, input delayed: $y\big(u(t-T), t\big) = u(t-T) - 0.5u(t+1-T)$.
- Left side, output delayed (replace every t by $t-T$): $y\big(u(t), t-T\big) = u(t-T) - 0.5u(t-T+1)$.
- Identical for every u, t and $T > 0$ ⇒ **time-invariant**.

Linearity. Let $u_1(t) \to y_1(t)$, $u_2(t) \to y_2(t)$ and apply $\alpha_1 u_1(t) + \alpha_2 u_2(t)$:

$$\begin{aligned}&\alpha_1 u_1(t) + \alpha_2 u_2(t) - 0.5\big(\alpha_1 u_1(t+1) + \alpha_2 u_2(t+1)\big)\\ &\quad = \alpha_1 \underbrace{\big(u_1(t) - 0.5u_1(t+1)\big)}_{y_1(t)} + \alpha_2 \underbrace{\big(u_2(t) - 0.5u_2(t+1)\big)}_{y_2(t)} = \alpha_1 y_1(t) + \alpha_2 y_2(t).\end{aligned}$$

⇒ **linear**.

Interpretation: a system that "looks one second ahead" — perfectly linear and time-invariant, but it cannot be built as a real-time device.

1(b) $y(t) = u(2t)$

Causality. The read time is $2t$. For $t > 0$, $2t > t$: the output needs future input values. (For $t < 0$ it reads the past, but Definition 1.3 must hold for all $T > 0$.)

Counterexample. $T = 1$, $u_1(t) = 0$ for all t, $u_2(t) = 0$ for $t \le 1$ and $u_2(t) = 1$ for $t > 1$. Then $u_1(t) = u_2(t)\ \forall t \le 1$. At $t = 0.75 \le 1$: $y_1(0.75) = u_1(1.5) = 0$, $y_2(0.75) = u_2(1.5) = 1$ ⇒ **not causal**.

Time-invariance. Careful use of the "replace t" rule (Concepts §1.2):

- Right side, input delayed: the system reads its input at time $2t$, and the input is now $u(t-T)$, so $y\big(u(t-T), t\big) = u(2t-T)$.
- Left side, output delayed: $y\big(u(t), t-T\big) = u\big(2(t-T)\big) = u(2t-2T)$.
- These differ: the delay at the output is $T/2$ instead of T (the compression also compresses the delay).

Counterexample. $u(t)$ = unit step ($u(t) = 1$ for $t \ge 0$, else 0), $T = 1$, $t = 0.75$: the right side gives $u(0.5) = 1$, the left side gives $u(-0.5) = 0$ ⇒ **time-variant**.

Linearity. $\alpha_1 u_1(t) + \alpha_2 u_2(t)$ produces $\alpha_1 u_1(2t) + \alpha_2 u_2(2t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$ ⇒ **linear**. (Changing the time argument never destroys linearity.)

Interpretation: playing a recording at double speed. It is linear, but a 1 s delay of the recording shows up as only 0.5 s at the output, and to play at double speed in real time you would need the future of the recording.

1(c) $y(t) = A\cos(2\pi f_0 t + u(t))$

This is **phase modulation**: the input sets the phase of a carrier. (A is the scalar amplitude here, not a matrix.)

Causality. Memoryless: $y(t)$ uses u only at the same instant t ⇒ **causal**. (If $u_1(t) = u_2(t)\ \forall t \le T$, then $y_1(t) = A\cos(2\pi f_0 t + u_1(t)) = A\cos(2\pi f_0 t + u_2(t)) = y_2(t)\ \forall t \le T$.)

Time-invariance.

- Right side: $y\big(u(t-T),t\big) = A\cos\big(2\pi f_0 t + u(t-T)\big)$ — the carrier term $2\pi f_0 t$ is *not* shifted, because it is part of the system, not of the input.
- Left side: $y\big(u(t),t-T\big) = A\cos\big(2\pi f_0(t-T) + u(t-T)\big)$.

Counterexample. $u(t) = 0$ for all t, $T = \frac{1}{4f_0}$ (a quarter period), $t = 0$: the right side gives $A\cos 0 = A$; the left side gives $A\cos(-\pi/2) = 0$. Different ⇒ **time-variant**. The explicit t in the carrier is the red flag (Concepts §3.3).

Linearity. Zero-in-zero-out test (Concepts §2.3): $u(t) = 0$ gives $y(t) = A\cos(2\pi f_0 t) \not\equiv 0$ ⇒ **not linear**. (Homogeneity also fails in general: $2u(t)$ produces $A\cos(2\pi f_0 t + 2u(t)) \neq 2A\cos(2\pi f_0 t + u(t))$; the input sits inside the cosine.)

1(d) $y(t) = u(t)\cos(2\pi f_0 t + \phi)$

This is **amplitude modulation** (a mixer): the input multiplies a carrier.

Causality. Memoryless ⇒ **causal**.

Time-invariance.

- Right side: $y\big(u(t-T),t\big) = u(t-T)\cos(2\pi f_0 t + \phi)$.
- Left side: $y\big(u(t),t-T\big) = u(t-T)\cos\big(2\pi f_0(t-T) + \phi\big)$.

Counterexample. $u(t) = 1$ for all t, $T = \frac{1}{4f_0}$, $t = -\frac{\phi}{2\pi f_0}$ (so that $2\pi f_0 t + \phi = 0$): the right side gives $\cos 0 = 1$, the left side gives $\cos(-\pi/2) = 0$ ⇒ **time-variant**.

Linearity. The factor $\cos(2\pi f_0 t + \phi)$ is a known function of time that does not depend on u:

$$\big(\alpha_1 u_1(t) + \alpha_2 u_2(t)\big)\cos(2\pi f_0 t + \phi) = \alpha_1 u_1(t)\cos(2\pi f_0 t + \phi) + \alpha_2 u_2(t)\cos(2\pi f_0 t + \phi) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

⇒ **linear** (a linear, time-variant system).

Compare (c) and (d): both contain a carrier with an explicit t, so both are time-variant. In (d) the input multiplies the cosine (linear); in (c) it sits inside the cosine (nonlinear).

Summary table and special cases

	causal	time-invariant	linear
a) $y(t) = u(t) - 0.5u(t+1)$	N	Y	Y
b) $y(t) = u(2t)$	N	N	Y
c) $y(t) = A\cos(2\pi f_0 t + u(t))$	Y	N	N
d) $y(t) = u(t)\cos(2\pi f_0 t + \phi)$	Y	N	Y

Special parameter values. If $f_0 = 0$, the explicit time dependence disappears: (c) becomes $A\cos(u(t))$ and (d) becomes $u(t)\cos\phi$, both time-invariant; (c) stays nonlinear unless $A = 0$ (then $y(t) = 0$, which is trivially linear and time-invariant).

Extra (not asked): all four are BIBO stable: $|y| \leq 1.5M$, $|y| \leq M$, $|y| \leq |A|$ and $|y| \leq M$ respectively.

Check: `python/task1_system_properties.py` tests all three properties numerically on random signals and reproduces the table and both counterexamples.

Task 2a — Linearization

$$\dot{x}_1(t) = -\sin x_1(t), \qquad \dot{x}_2(t) = -x_2(t) + \cos x_1(t), \qquad \text{i.e.} \qquad \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} f_1(\boldsymbol{x}) \\ f_2(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} -\sin x_1 \\ -x_2 + \cos x_1 \end{bmatrix}.$$

The system is autonomous (no input), so the linear model is $\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{x}) = \boldsymbol{A}\Delta\boldsymbol{x}$.

Step 1 — Equilibrium points (Definition 1.6)

Solve $\boldsymbol{f}(\boldsymbol{x}_0) = 0$ (Concepts §8.2). The first component contains only x_1:

$$-\sin x_1 = 0 \iff x_1 = k\pi, \quad k \in \mathbb{Z}.$$

Substitute into the second: $-x_2 + \cos(k\pi) = 0 \iff x_2 = \cos(k\pi) = (-1)^k$.

$$\boxed{\boldsymbol{x}_0 = \begin{bmatrix} k\pi \\ (-1)^k \end{bmatrix}, \qquad k \in \mathbb{Z}}$$

So, as for the pendulum of slide 18, there is an infinite number of equilibrium points, in two families:

- k even: $(0, 1), (\pm 2\pi, 1), (\pm 4\pi, 1), \ldots$
- k odd: $(\pm\pi, -1), (\pm 3\pi, -1), \ldots$

Check $k = 1$: $f_1 = -\sin\pi = 0$, $f_2 = -(-1) + \cos\pi = 1 - 1 = 0$ ✓.

Official solution: the official sheet restricts the angle to one turn, $x_1 \in [0, 2\pi)$. In that range only $k = 0$ and $k = 1$ remain, i.e. the two equilibrium points

$$\boldsymbol{x}^{*(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{x}^{*(2)} = \begin{bmatrix} \pi \\ -1 \end{bmatrix},$$

which are exactly the "k even" and "k odd" representatives above. (The official sheet writes the second one as $(k\pi, -1)$; note that $x_2 = -1$ holds only for **odd** k — for even k the general formula gives $x_2 = +1$.) Both presentations lead to the same two linear models in Step 4.

Step 2 — Jacobian matrix (symbolic)

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} -\cos x_1 & 0 \\ -\sin x_1 & -1 \end{bmatrix}.$$

($\frac{\partial}{\partial x_1}(-\sin x_1) = -\cos x_1$; f_1 does not contain x_2; $\frac{\partial}{\partial x_1}\cos x_1 = -\sin x_1$; $\frac{\partial}{\partial x_2}(-x_2) = -1$.)

Step 3 — Evaluate at the equilibrium points

Use $\cos(k\pi) = (-1)^k$, $\sin(k\pi) = 0$:

$$\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}_0} = \begin{bmatrix} -(-1)^k & 0 \\ 0 & -1 \end{bmatrix} \quad \Longrightarrow \quad \boldsymbol{A}_{\text{even}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \boldsymbol{A}_{\text{odd}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Step 4 — Linearized models

With $\Delta \boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$:

$$\boxed{k \text{ even}: \ \frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\Delta x_1 \\ \Delta x_2\end{bmatrix} = \begin{bmatrix}-1 & 0 \\ 0 & -1\end{bmatrix}\begin{bmatrix}\Delta x_1 \\ \Delta x_2\end{bmatrix}, \qquad k \text{ odd}: \ \frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\Delta x_1 \\ \Delta x_2\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix}\begin{bmatrix}\Delta x_1 \\ \Delta x_2\end{bmatrix}}$$

where, e.g. for $k = 0$, $\Delta x_1 = x_1$, $\Delta x_2 = x_2 - 1$; for $k = 1$, $\Delta x_1 = x_1 - \pi$, $\Delta x_2 = x_2 + 1$.

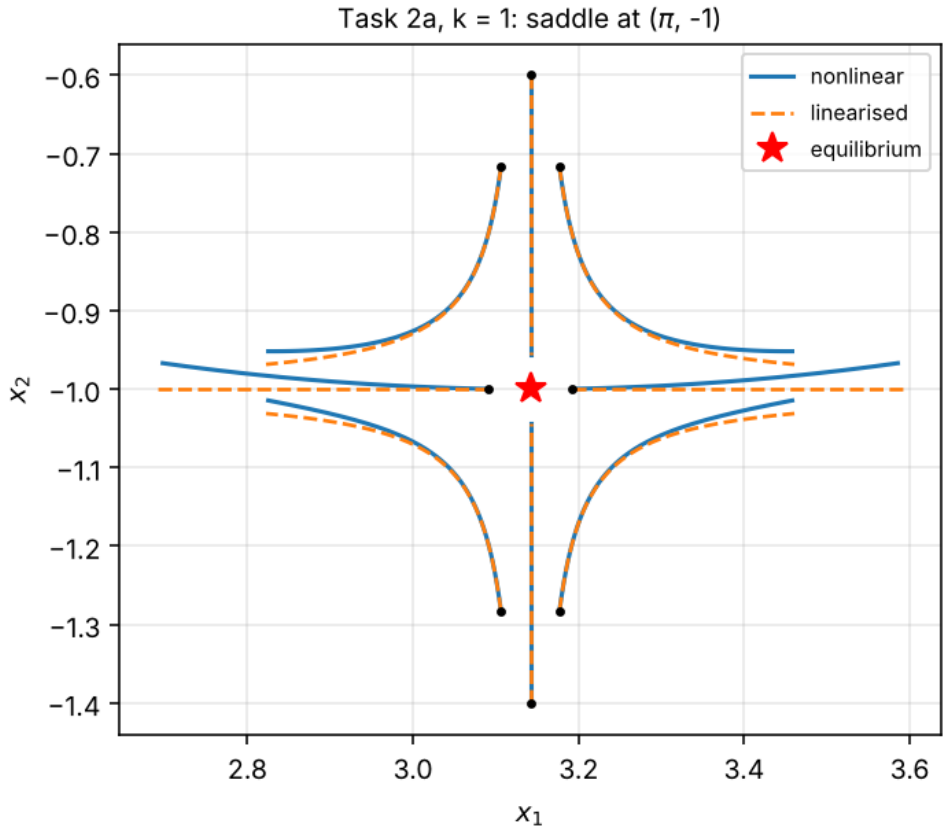
Step 5 — Interpretation (Concepts §10)

The matrices are diagonal, so the eigenvalues are the diagonal entries.

- k even: $\lambda = -1, -1 \Rightarrow$ both negative ⇒ **locally asymptotically stable** (stable node). Small deviations decay like e^{-t}.
- k odd: $\lambda = +1, -1 \Rightarrow$ **saddle, unstable**. Deviations in x_1 grow like e^{t}; deviations in x_2 decay.

The structure is the same as for the pendulum: $\dot{x}_1 = -\sin x_1$ pushes x_1 towards the nearest even multiple of π and away from odd multiples; x_2 simply follows $\cos x_1$ with a first-order lag.

Check: `python/task2_linearization.py` (symbolic Jacobians for $k = -2, \ldots, 2$); figures `task2a_k0_stable.png` and `task2a_k1_saddle.png` overlay nonlinear and linearized trajectories.



Task 2b — Linearization of the second nonlinear system

$$\begin{aligned}\dot{x}_1(t) &= x_1(t)^2 + x_2(t)^2 + x_2(t)\cos x_1(t) &&=: f_1(\boldsymbol{x}),\\ \dot{x}_2(t) &= (1+x_1(t))x_1(t) + (1+x_2(t))x_2(t) + x_1(t)\sin x_2(t) &&=: f_2(\boldsymbol{x}).\end{aligned}$$

Multiplied out: $f_2 = x_1 + x_1^2 + x_2 + x_2^2 + x_1 \sin x_2$.

Step 1 — Equilibrium points

We need $\boldsymbol{f}(\boldsymbol{x}_0) = 0$:

$$\text{(I)}\quad x_1^2 + x_2^2 + x_2\cos x_1 = 0, \qquad \text{(II)}\quad x_1^2 + x_2^2 + x_1 + x_2 + x_1\sin x_2 = 0.$$

Obvious candidate. $\boldsymbol{x} = 0$: (I) $= 0$, (II) $= 0$ ✓.

Look along $x_1 = 0$**.** Then (I) becomes $x_2^2 + x_2 = x_2(x_2+1) = 0$ and (II) becomes $x_2^2 + x_2 = 0$ — the same equation. So $x_2 = 0$ or $x_2 = -1$:

$$(0,0) \quad \text{and} \quad (0,-1).$$

Check $(0,-1)$*:* (I) $= 0 + 1 + (-1)\cos 0 = 0$ ✓; (II) $= 0 + 1 + 0 - 1 + 0 = 0$ ✓.

Are there others? We show there are none (Concepts §8.2, steps 4–5).

(i) Confine the solutions. From (I), $x_1^2 + x_2^2 = -x_2\cos x_1 \le |x_2|$ (because $|\cos| \le 1$). Hence $x_2^2 \le |x_2|$, i.e. $|x_2| \le 1$, and $x_1^2 \le |x_2| - x_2^2 \le \frac{1}{4}$ (the maximum of $s - s^2$ is $\frac{1}{4}$), i.e. $|x_1| \le \frac{1}{2}$. Then $\cos x_1 > 0$, and since $-x_2\cos x_1 = x_1^2 + x_2^2 \ge 0$ we get $x_2 \le 0$. So every equilibrium lies in $|x_1| \le \frac{1}{2}$, $-1 \le x_2 \le 0$.

(ii) Combine the equations. Subtract (I) from (II):

$$x_1 + x_2 + x_1\sin x_2 - x_2\cos x_1 = 0 \quad\Longleftrightarrow\quad x_1\,(1+\sin x_2) = -x_2\,(1-\cos x_1). \qquad \text{(III)}$$

In the region from (i): $1 + \sin x_2 \ge 1 - \sin 1 \approx 0.159 > 0$, $-x_2 \ge 0$, $1 - \cos x_1 \ge 0$. So the right side is ≥ 0, hence $x_1 \ge 0$.

(iii) Rule out $x_1 > 0$*.* Suppose $x_1 > 0$. Using $1 - \cos x_1 \le x_1^2/2$ and $|x_2| \le 1$ in (III): $x_1(1 + \sin x_2) \le x_1^2/2$, so $1 + \sin x_2 \le x_1/2 \le \frac{1}{4}$, which forces $\sin x_2 \le -\frac{3}{4}$, i.e. $x_2 \le \arcsin(-0.75) \approx -0.848$. Feeding this back into $x_1^2 \le |x_2|(1-|x_2|)$ (from (i)) gives $x_1^2 \le 0.848\cdot 0.152 \approx 0.129$, so $x_1 \le 0.359$; then $1 + \sin x_2 \le 0.180$, $\sin x_2 \le -0.820$, $x_2 \le -0.961$, so $x_1^2 \le 0.961 \cdot 0.039 \approx 0.037$, $x_1 \le 0.194$; then $1 + \sin x_2 \le 0.097$, i.e. $\sin x_2 \le -0.903$. (Each time we use that $s(1-s)$ decreases for $s > \frac{1}{2}$, so its largest value on $[|x_2|_{\min}, 1]$ is at the left end.) But on $[-1, 0]$, $\sin x_2 \ge \sin(-1) \approx -0.841$. Contradiction. Hence $x_1 = 0$, and we already solved that case.

equilibrium points: $\boldsymbol{x}_0 = \begin{bmatrix}0\\0\end{bmatrix}$ and $\boldsymbol{x}_0 = \begin{bmatrix}0\\-1\end{bmatrix}$

(The script additionally runs a dense numerical root search on $[-3,3]^2$ and finds exactly these two points.)

Step 2 — Jacobian matrix (symbolic)

Differentiate term by term (Concepts §9.3 table):

$$\frac{\partial f_1}{\partial x_1} = 2x_1 - x_2 \sin x_1, \quad \frac{\partial f_1}{\partial x_2} = 2x_2 + \cos x_1, \quad \frac{\partial f_2}{\partial x_1} = 1 + 2x_1 + \sin x_2, \quad \frac{\partial f_2}{\partial x_2} = 1 + 2x_2 + x_1 \cos x_2.$$

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} 2x_1 - x_2 \sin x_1 & 2x_2 + \cos x_1 \\ 1 + 2x_1 + \sin x_2 & 1 + 2x_2 + x_1 \cos x_2 \end{bmatrix}.$$

Official solution: the bottom-right entry is printed there as $1 + 2x_2 + x_1 \cos x_1$ — a typo: $\frac{\partial}{\partial x_2}(x_1 \sin x_2) = x_1 \cos x_2$. Both equilibria have $x_1 = 0$, so the term vanishes and the numerical matrices are identical. The official proof that no other equilibria exist uses a different inequality argument (completing the square in (I), then concavity of sin); it reaches the same conclusion $x_1 = 0$.

Step 3 — Evaluate at $\boldsymbol{x}_0 = [0,\ 0]^\top$

$\sin 0 = 0$, $\cos 0 = 1$:

$$\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(0,0)} = \begin{bmatrix} 0-0 & 0+1 \\ 1+0+0 & 1+0+0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \boxed{\frac{\mathrm{d}}{\mathrm{d}t}(\Delta \boldsymbol{x}) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \Delta \boldsymbol{x}, \quad \Delta \boldsymbol{x} = \boldsymbol{x}}$$

Eigenvalues: $\operatorname{tr} \boldsymbol{A} = 1$, $\det \boldsymbol{A} = 0 \cdot 1 - 1 \cdot 1 = -1$, so $\lambda^2 - \lambda - 1 = 0$, $\lambda = \frac{1 \pm \sqrt{5}}{2} \approx 1.618,\ -0.618$. $\det \boldsymbol{A} < 0 \Rightarrow$ **saddle, unstable**.

Step 4 — Evaluate at $\boldsymbol{x}_0 = [0,\ -1]^\top$

$\sin 0 = 0$, $\cos 0 = 1$, $\sin(-1) = -\sin 1$, $\cos(-1) = \cos 1$:

$$\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(0,-1)} = \begin{bmatrix} 0 - (-1) \cdot 0 & -2 + 1 \\ 1 + 0 - \sin 1 & 1 - 2 + 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 - \sin 1 & -1 \end{bmatrix} \approx \begin{bmatrix} 0 & -1 \\ 0.1585 & -1 \end{bmatrix}.$$

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(\Delta \boldsymbol{x}) = \begin{bmatrix} 0 & -1 \\ 1 - \sin 1 & -1 \end{bmatrix} \Delta \boldsymbol{x}, \qquad \Delta \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 + 1 \end{bmatrix}}$$

Eigenvalues: $\operatorname{tr} \boldsymbol{A} = -1$, $\det \boldsymbol{A} = 0 \cdot (-1) - (-1)(1 - \sin 1) = 1 - \sin 1 \approx 0.1585$:

$$\lambda = \frac{-1 \pm \sqrt{1 - 4(1 - \sin 1)}}{2} = \frac{-1 \pm \sqrt{4 \sin 1 - 3}}{2} \approx -0.198,\ -0.802.$$

$\operatorname{tr} \boldsymbol{A} < 0$, $\det \boldsymbol{A} > 0 \Rightarrow$ both negative real $\Rightarrow$ **stable node, locally asymptotically stable**.

Check: `python/task2_linearization.py`; figure `task2b_stable_node.png`.

Task 2c — State space and linearization of the I/O relation

$$\ddot{y}(t) + (3 + \dot{y}(t)^2)\,\dot{y}(t) + (1 + y(t) + y(t)^2)\,u(t) = 0, \qquad \boldsymbol{x} = [x_1,\ x_2]^\top = [y,\ \dot{y}]^\top.$$

Step 1 — Nonlinear state-space representation (Concepts §7.2)

Solve for the highest derivative: $\ddot{y} = -(3+\dot{y}^2)\dot{y} - (1+y+y^2)u$. With $\dot{x}_1 = \dot{y} = x_2$ and $\dot{x}_2 = \ddot{y}$:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, u) = \begin{bmatrix} x_2 \\ -(3 + x_2^2)\,x_2 - (1 + x_1 + x_1^2)\,u \end{bmatrix}, \qquad y = g(\boldsymbol{x}, u) = x_1}$$

Multiplied out: $f_2 = -3x_2 - x_2^3 - u - x_1 u - x_1^2 u$.

Step 2 — Check that the origin is an equilibrium point

$f_1(0,0) = 0$, $f_2(0,0) = -(3+0)\cdot 0 - (1+0+0)\cdot 0 = 0$ ✓. So $(\boldsymbol{x}^*, u^*) = (0,0)$ is an equilibrium point and $y^* = g(0,0) = 0$.

(Bonus: all equilibrium points.) $f_1 = 0 \Rightarrow x_2 = 0$; then $f_2 = -(1 + x_1 + x_1^2)u = 0$. The factor $1 + x_1 + x_1^2 = (x_1 + \frac{1}{2})^2 + \frac{3}{4} > 0$ never vanishes, so $u^* = 0$. Hence the equilibrium points are $\boldsymbol{x}^* = [x_1^*,\ 0]^\top$ with **any** x_1^* and $u^* = 0$ — a whole line. Constant nonzero inputs have no equilibrium.

Step 3 — Jacobian matrices

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} 0 & 1 \\ -(1 + 2x_1)\,u & -3 - 3x_2^2 \end{bmatrix}, \qquad \frac{\partial \boldsymbol{f}}{\partial u} = \begin{bmatrix} 0 \\ -(1 + x_1 + x_1^2) \end{bmatrix}, \qquad \frac{\partial g}{\partial \boldsymbol{x}} = [1 \quad 0], \qquad \frac{\partial g}{\partial u} = 0.$$

(For $\frac{\partial f_2}{\partial x_2}$: $\frac{\partial}{\partial x_2}(-3x_2 - x_2^3) = -3 - 3x_2^2$. For $\frac{\partial f_2}{\partial x_1}$: $\frac{\partial}{\partial x_1}(-(1 + x_1 + x_1^2)u) = -(1 + 2x_1)u$.)

Step 4 — Evaluate at $(\boldsymbol{x}^*, u^*) = (0,0)$

$$\boxed{\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}^*, u^*)} = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}, \quad \boldsymbol{B} = \left.\frac{\partial \boldsymbol{f}}{\partial u}\right|_{(\boldsymbol{x}^*, u^*)} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad \boldsymbol{C} = [1 \quad 0], \quad \boldsymbol{D} = 0}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{x}) = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}\Delta\boldsymbol{x} + \begin{bmatrix} 0 \\ -1 \end{bmatrix}\Delta u, \qquad \Delta y = [1 \quad 0]\,\Delta\boldsymbol{x}.$$

Here $\Delta\boldsymbol{x} = \boldsymbol{x}$, $\Delta u = u$, $\Delta y = y$ because the operating point is the origin. Note that $\frac{\partial f_2}{\partial x_1} = -(1 + 2x_1)u$ vanishes because $u^* = 0$: around the origin the dependence of the gain on y is a second-order effect.

Equivalent linear I/O relation. $\Delta\dot{x}_1 = \Delta x_2$, $\Delta\dot{x}_2 = -3\Delta x_2 - \Delta u$, $\Delta y = \Delta x_1$ give

$$\Delta\ddot{y}(t) + 3\,\Delta\dot{y}(t) + \Delta u(t) = 0.$$

This is exactly what you get by dropping all products of small quantities ($\dot{y}^3$, yu, y^2u) in the original equation — a good sanity check.

Step 5 — Interpretation

$\boldsymbol{A}$ is upper triangular, so its eigenvalues are 0 and -3. The eigenvalue 0 reflects the line of equilibria: shifting y by a constant (with $u = 0$, $\dot{y} = 0$) gives another equilibrium, so a perturbation of y neither decays nor grows. Because one eigenvalue has zero real part, the linearization alone cannot decide the stability of the nonlinear system (Concepts §10.3).

Check: `python/task2_linearization.py` computes $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ and the eigenvalues symbolically.

Task 3 — Evolution of a discrete-time system

$$\boldsymbol{x}[k+1] = \underbrace{\begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x}[k] + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\boldsymbol{B}} u[k], \qquad \boldsymbol{y}[k] = \boldsymbol{x}[k] \quad (\boldsymbol{C} = \boldsymbol{I},\ \boldsymbol{D} = 0), \qquad k \in \mathbb{Z}.$$

Preparation — eigenvalues and eigenvectors of $\boldsymbol{A}$ (Concepts §12.1)

tr $\boldsymbol{A} = -3$, det $\boldsymbol{A} = 0 \cdot (-3) - (-2) \cdot 1 = 2$:

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0 \implies \lambda_1 = -1, \quad \lambda_2 = -2.$$

Each eigenvector solves $(\lambda_i \boldsymbol{I} - \boldsymbol{A})\boldsymbol{v} = 0$ — the same matrix as in the characteristic equation. (Using $\boldsymbol{A} - \lambda_i \boldsymbol{I}$ instead flips every sign and gives the same eigenvectors.)

- $\lambda_1 = -1$: $\lambda_1 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} -1-0 & 0+2 \\ 0-1 & -1+3 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix}$, the row gives $-v_1 + 2v_2 = 0 \Rightarrow \boldsymbol{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Check: $\boldsymbol{A}\boldsymbol{v}_1 = \begin{bmatrix} 0-2 \\ 2-3 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix} = -\boldsymbol{v}_1$ ✓.
- $\lambda_2 = -2$: $\lambda_2 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} -2-0 & 0+2 \\ 0-1 & -2+3 \end{bmatrix} = \begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix}$, the row gives $-v_1 + v_2 = 0 \Rightarrow \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Check: $\boldsymbol{A}\boldsymbol{v}_2 = \begin{bmatrix} -2 \\ 1-3 \end{bmatrix} = -2\boldsymbol{v}_2$ ✓.

Notice: $\boldsymbol{B} = \boldsymbol{v}_2$. The input enters exactly along the eigenvector of $\lambda_2 = -2$. This will matter in (c) and (d).

3(a) — Iteration and closed form ($u[k] = 0$ for all k, $\boldsymbol{x}[0] = [1,\ 0]^\top$)

Iteration ($\boldsymbol{x}[k+1] = \boldsymbol{A}\boldsymbol{x}[k]$, row times column):

$$\boldsymbol{x}[1] = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + (-2) \cdot 0 \\ 1 \cdot 1 + (-3) \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$\boldsymbol{x}[2] = \boldsymbol{A} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0-2 \\ 0-3 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}, \qquad \boldsymbol{x}[3] = \boldsymbol{A} \begin{bmatrix} -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 0+6 \\ -2+9 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}.$$

Closed form, method 1 (decompose $\boldsymbol{x}[0]$, Concepts §12.2). Write $\boldsymbol{x}[0] = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \iff \begin{cases} 2c_1 + c_2 = 1 \\ c_1 + c_2 = 0 \end{cases} \iff c_1 = 1,\ c_2 = -1.$$

Since $\boldsymbol{A}^k\boldsymbol{v}_i = \lambda_i^k\boldsymbol{v}_i$:

$$\boxed{\boldsymbol{x}[k] = (-1)^k\begin{bmatrix}2\\1\end{bmatrix} - (-2)^k\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2(-1)^k-(-2)^k\\(-1)^k-(-2)^k\end{bmatrix}}$$

Check: $k=0$: $[2-1,\ 1-1]^\top = [1,0]^\top$ ✓; $k=1$: $[-2+2,\ -1+2]^\top = [0,1]^\top$ ✓; $k=2$: $[2-4,\ 1-4]^\top = [-2,-3]^\top$ ✓; $k=3$: $[-2+8,\ -1+8]^\top = [6,7]^\top$ ✓.

Closed form, method 2 (general $\boldsymbol{A}^k$, Concepts §12.3). $\boldsymbol{V} = [\boldsymbol{v}_1\ \boldsymbol{v}_2] = \begin{bmatrix}2 & 1\\1 & 1\end{bmatrix}$, $\det\boldsymbol{V} = 1$, $\boldsymbol{V}^{-1} = \begin{bmatrix}1 & -1\\-1 & 2\end{bmatrix}$, $\boldsymbol{\Lambda} = \mathrm{diag}(-1,-2)$:

$$\boldsymbol{A}^k = \boldsymbol{V}\boldsymbol{\Lambda}^k\boldsymbol{V}^{-1} = (-1)^k\begin{bmatrix}2\\1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix} + (-2)^k\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}-1 & 2\end{bmatrix} = (-1)^k\begin{bmatrix}2 & -2\\1 & -1\end{bmatrix} + (-2)^k\begin{bmatrix}-1 & 2\\-1 & 2\end{bmatrix},$$

$$\boxed{\boldsymbol{A}^k = \begin{bmatrix}2(-1)^k-(-2)^k & -2(-1)^k+2(-2)^k\\(-1)^k-(-2)^k & -(-1)^k+2(-2)^k\end{bmatrix}}$$

Check: $k=0$ gives $\boldsymbol{I}$; $k=1$ gives $\begin{bmatrix}-2+2 & 2-4\\-1+2 & 1-4\end{bmatrix} = \boldsymbol{A}$ ✓. Multiplying by $\boldsymbol{x}[0] = [1,0]^\top$ picks the first column — the same result as method 1.

3(b) — Convergence and internal stability

For $\boldsymbol{x}[0] = [1,0]^\top$: $\boldsymbol{x}[k]$ contains $-(-2)^k[1,1]^\top$, whose size is $2^k \to \infty$ with alternating sign $(1,-2,4,-8,\ldots)$. The other part $(-1)^k[2,1]^\top$ stays bounded but keeps flipping sign. So $\|\boldsymbol{x}[k]\| \to \infty$: **the state does not converge; it diverges with alternating sign** (as $\boldsymbol{x}[3] = [6,7]^\top$ already suggests).

For arbitrary $\boldsymbol{x}[0] \in \mathbb{R}^2$: decompose $\boldsymbol{x}[0] = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$ (always possible, $\boldsymbol{v}_1, \boldsymbol{v}_2$ are a basis). Writing $\boldsymbol{x}[0] = [a,\ b]^\top$:

$$\begin{bmatrix}a\\b\end{bmatrix} = c_1\begin{bmatrix}2\\1\end{bmatrix} + c_2\begin{bmatrix}1\\1\end{bmatrix} \iff \begin{cases}2c_1+c_2 = a\\c_1+c_2 = b\end{cases} \iff c_1 = a-b,\quad c_2 = 2b-a$$

(subtract the second equation from the first to get c_1; then $c_2 = b - c_1$). *Check:* $(a-b)[2,1]^\top + (2b-a)[1,1]^\top = [a,\ b]^\top$ ✓. Hence

$$\boldsymbol{x}[k] = (a-b)(-1)^k\begin{bmatrix}2\\1\end{bmatrix} + (2b-a)(-2)^k\begin{bmatrix}1\\1\end{bmatrix}.$$

Case	Behaviour of $\boldsymbol{x}[k]$
$c_2 \neq 0$ (i.e. $\boldsymbol{x}[0]$ not a multiple of $[2,1]^\top$)	diverges ($\lvert\lambda_2\rvert = 2 > 1$)
$c_2 = 0$, $c_1 \neq 0$ ($\boldsymbol{x}[0] = c_1[2,1]^\top$)	bounded but **not convergent**: alternates between $c_1[2,1]^\top$ and $-c_1[2,1]^\top$ ($\lambda_1 = -1$ on the unit circle)

Case	Behaviour of $\boldsymbol{x}[k]$
$\boldsymbol{x}[0] = 0$	stays 0

For convergence both coefficients must vanish: $a - b = 0$ and $2b - a = 0$, i.e. $a = b = 0$. Hence $\boldsymbol{x}[k]$ **converges only for** $\boldsymbol{x}[0] = 0$ **—** the system is **not internally asymptotically stable** (the official solution's wording). In the language of Concepts §13.2: the largest eigenvalue magnitude is $\max|\lambda_i| = 2 > 1$, so the system is **not internally stable (unstable)**. Even without λ_2, the eigenvalue $\lambda_1 = -1$ alone would prevent asymptotic stability.

3(c) — Closed form with input

From Concepts §11.3:

$$\boldsymbol{x}[k] = \boldsymbol{A}^k \boldsymbol{x}[0] + \sum_{i=0}^{k-1} \boldsymbol{A}^{k-1-i} \boldsymbol{B}\, u[i], \qquad k \geq 1.$$

Because $\boldsymbol{B} = \boldsymbol{v}_2$ is an eigenvector with $\lambda_2 = -2$, $\boldsymbol{A}^j \boldsymbol{B} = (-2)^j \boldsymbol{B}$ (Concepts §12.5), so the sum simplifies dramatically:

$$\boxed{\boldsymbol{x}[k] = \boldsymbol{A}^k \boldsymbol{x}[0] + \Big(\sum_{i=0}^{k-1} (-2)^{k-1-i}\, u[i] \Big) \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{A}^k \text{ as in 3(a)}}$$

and $\boldsymbol{y}[k] = \boldsymbol{x}[k]$. *Check* $k = 1$*:* $\boldsymbol{x}[1] = \boldsymbol{A}\boldsymbol{x}[0] + (-2)^0 u[0]\boldsymbol{B} = \boldsymbol{A}\boldsymbol{x}[0] + \boldsymbol{B}u[0]$ ✓. *Check* $k = 2$*:* $\boldsymbol{A}^2\boldsymbol{x}[0] + (-2u[0] + u[1])\boldsymbol{B}$, and directly $\boldsymbol{x}[2] = \boldsymbol{A}\boldsymbol{x}[1] + \boldsymbol{B}u[1] = \boldsymbol{A}^2\boldsymbol{x}[0] + \boldsymbol{A}\boldsymbol{B}u[0] + \boldsymbol{B}u[1] = \boldsymbol{A}^2\boldsymbol{x}[0] - 2\boldsymbol{B}u[0] + \boldsymbol{B}u[1]$ ✓.

For $\boldsymbol{x}[0] = [1, 0]^\top$ this reads componentwise:

$$\boldsymbol{x}[k] = \begin{bmatrix} 2(-1)^k - (-2)^k + s_k \\ (-1)^k - (-2)^k + s_k \end{bmatrix}, \qquad s_k = \sum_{i=0}^{k-1} (-2)^{k-1-i} u[i].$$

Interpretation: the input only ever excites the $\lambda = -2$ mode; the $\lambda = -1$ mode is untouched by u (the difference $x_1[k] - x_2[k] = (-1)^k (x_1[0] - x_2[0])$ evolves independently of the input).

3(d) — Impulse response and BIBO stability

With $\boldsymbol{x}[0] = 0$ and $u[k] = \delta[k]$, only the $i = 0$ term of the sum survives:

$$\boldsymbol{x}[0] = 0, \qquad \boldsymbol{x}[k] = \boldsymbol{A}^{k-1}\boldsymbol{B} = (-2)^{k-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (k \geq 1), \qquad \boldsymbol{y}[k] = \boldsymbol{x}[k].$$

Values: $\boldsymbol{y}[1] = [1, 1]^\top$, $\boldsymbol{y}[2] = [-2, -2]^\top$, $\boldsymbol{y}[3] = [4, 4]^\top$, $\boldsymbol{y}[4] = [-8, -8]^\top$, ..., $|y_i[k]| = 2^{k-1} \to \infty$.

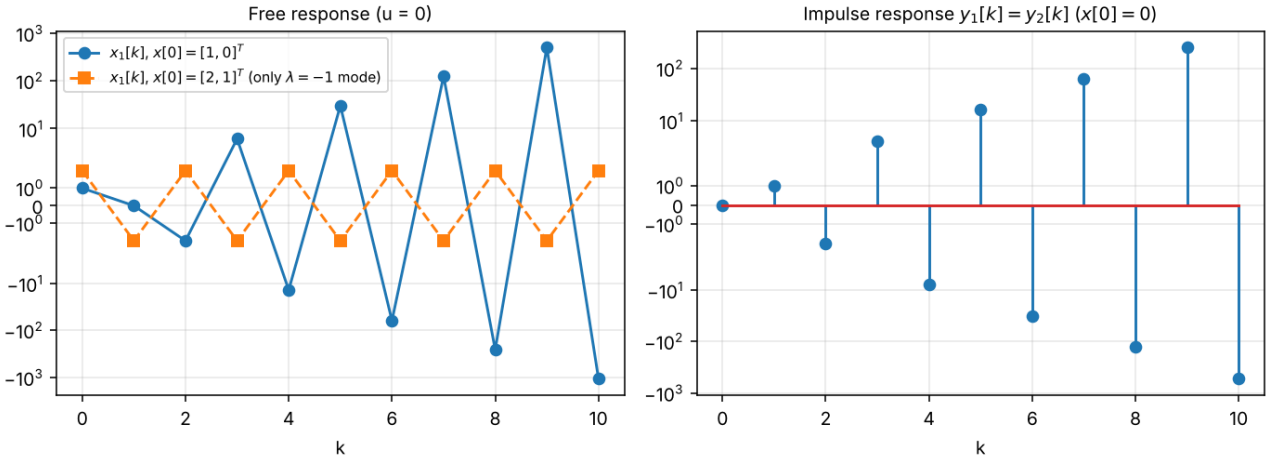
The input $\delta[k]$ is bounded ($|u[k]| \leq 1$), but the output is unbounded. **The output does not remain bounded as** $k \to \infty$ **⇒ the system is not BIBO stable.**

Equivalently (as in the official solution): an LTI system is BIBO stable exactly when its impulse response is **absolutely summable**. Here it is not:

$$\sum_{k=0}^{\infty}\|\boldsymbol{y}[k]\| = \left\|[1,\ 1]^\top\right\| \sum_{k=1}^{\infty} 2^{k-1} = \sqrt{2}\,(1+2+4+8+\dots) = \infty.$$

Link to (b). The unstable eigenvalue $\lambda_2 = -2$ is excited directly by the input (since $\boldsymbol{B} = \boldsymbol{v}_2$) and seen by the output ($\boldsymbol{C} = \boldsymbol{I}$), so internal instability shows up in the I/O behaviour. If $\boldsymbol{B}$ had been along $\boldsymbol{v}_1$ instead, the impulse response would be $(-1)^{k-1}\boldsymbol{v}_1$ — bounded, though not decaying — even though the system is still internally unstable. This is why internal stability and BIBO stability are different concepts (Concepts §13.3).

Check: `python/task3_discrete_time.py` verifies all numbers, both closed forms, the forced response against a simulation with random inputs, and plots the responses:



Summary of results

- **Task 1** (causal / time-invariant / linear): a) N Y Y; b) N N Y; c) Y N N; d) Y N Y.
- **Task 2a:** equilibrium points $\boldsymbol{x}_0 = [k\pi,\ (-1)^k]^\top$; $\boldsymbol{A} = \mathrm{diag}(-1,-1)$ for even k (stable), $\boldsymbol{A} = \mathrm{diag}(1,-1)$ for odd k (saddle).
- **Task 2b:** equilibrium points $[0,0]^\top$ and $[0,-1]^\top$ only; $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ (saddle, $\lambda = \frac{1\pm\sqrt{5}}{2}$) and $\boldsymbol{A} = \begin{bmatrix} 0 & -1 \\ 1-\sin 1 & -1 \end{bmatrix}$ (stable node, $\lambda \approx -0.198, -0.802$).
- **Task 2c:** $\dot{x}_1 = x_2$, $\dot{x}_2 = -(3+x_2^2)x_2 - (1+x_1+x_1^2)u$, $y = x_1$; at $(\boldsymbol{x}^*, u^*) = (0,0)$: $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $\boldsymbol{C} = [1\ 0]$, $\boldsymbol{D} = 0$, i.e. $\Delta\ddot{y} + 3\Delta\dot{y} + \Delta u = 0$.
- **Task 3:** $\boldsymbol{x}[1] = [0,1]^\top$, $\boldsymbol{x}[2] = [-2,-3]^\top$, $\boldsymbol{x}[3] = [6,7]^\top$; $\boldsymbol{x}[k] = (-1)^k[2,1]^\top - (-2)^k[1,1]^\top$; diverges, converges only for $\boldsymbol{x}[0] = 0$ (eigenvalues $-1, -2$); $\boldsymbol{x}[k] = \boldsymbol{A}^k\boldsymbol{x}[0] + \sum_{i<k}(-2)^{k-1-i}u[i]\,[1,1]^\top$; impulse response $(-2)^{k-1}[1,1]^\top$ is unbounded ⇒ not BIBO stable.